

MA 403 Real Analysis

Homework 1

1. Prove that $\sqrt{2} + \sqrt{3}$ is irrational.
2. If r is rational ($r \neq 0$) and x is irrational, prove that $r + x$ and rx are irrational.
3. Given any real number $x > 0$, prove that there is an irrational number between 0 and x .
4. Suppose x and y are real numbers and for each $\epsilon > 0$, $|x - y| \leq \epsilon$. Show that $x = y$.
5. Give an example of a bounded set S such that $\sup S$ is in S but $\inf S$ is not in S .
6. Suppose A and B are two subsets of $\mathbb{R}$ such that A is bounded from above and B is bounded from below. Show that the intersection $A \cap B$ is bounded from both above and below.
7. Let S be a (nonempty) set of real numbers such that $\sup S$ and $\inf S$ exist. Show that $\sup S$ and $\inf S$ are uniquely determined.
8. Let A and B be two sets of positive numbers which are bounded above, and let $a = \sup A$, $b = \sup B$. Let C be the set defined by

$$C = \{xy : x \in A \text{ and } y \in B\}$$

Prove that $ab = \sup C$.